GESTURE CONTROLLED WIRELESS 4WD ROBOTIC CAR

MINI-PROJECT PRESENTATION

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11. APPLICATIONS

1. Introduction & Project Overview

Intuitive Human-Machine Interfaces (HMI)
Standard controllers (joysticks, remote control pads) require physical contact, manual dexterity, and training. Gesture-based steering offers an intuitive, hands-free interface.

Gesture-Based Motion Mapping
Translates natural hand tilts into system command vectors. Tilting the hand forward, backward, left, or right coordinates physical direction changes in real-time.

ESP32/ESP-NOW Architecture
Using Espressif's connectionless protocol reduces transmission latency to a minimum, bypassing the handshaking delays and router pairing required in traditional Wi-Fi setups.

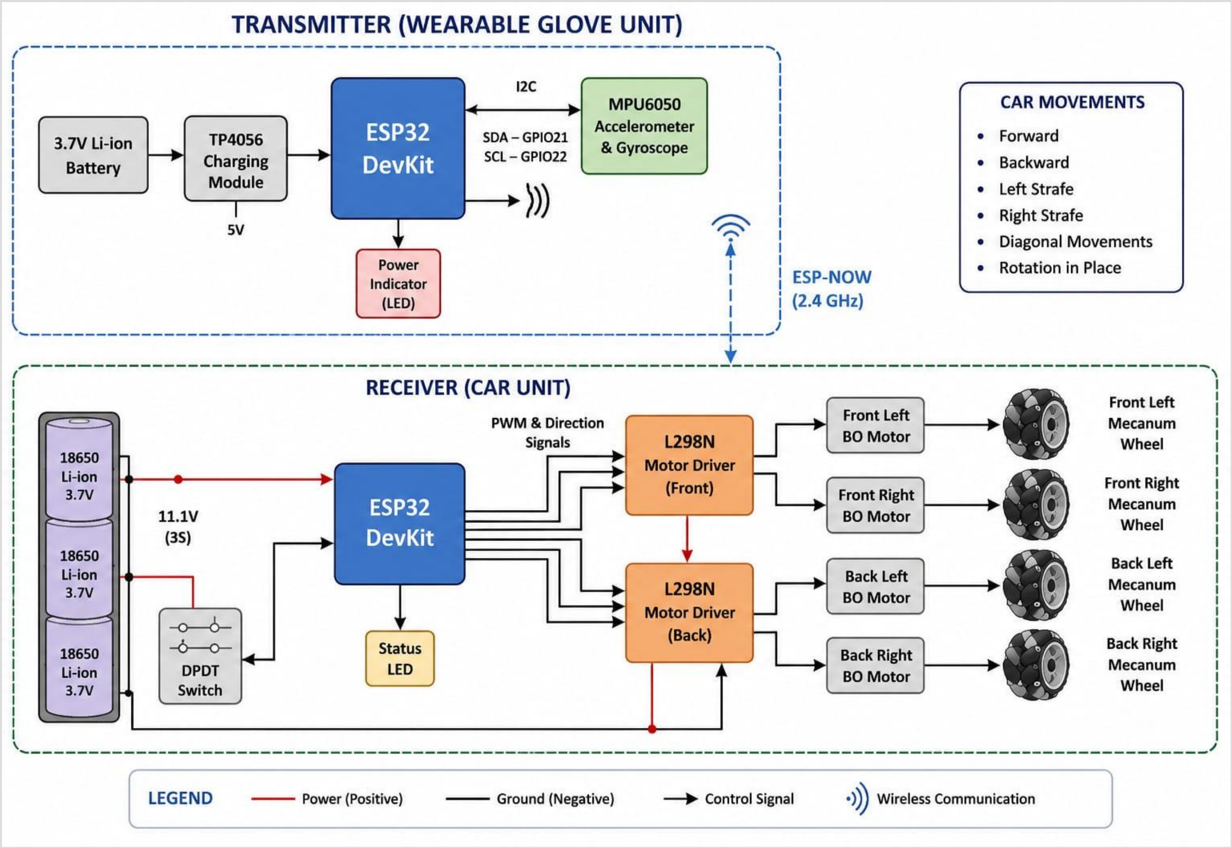
Real-World Application Scope
Highly applicable to heavy warehouse logistics, search-and-rescue assistance in hazardous areas, and assistive mobility interfaces for disabled individuals.

2. System Architecture Design

Transmitter Wearable Glove
Interfaced with the MPU6050 motion sensor. Tracks static gravity vectors to compute wrist orientation (Pitch and Roll).

ESP-NOW Direct Communication Link
Broadcasts structured packets over the 2.4 GHz band directly to the receiver's physical MAC address, minimizing delay.

Receiver Mobile Chassis
Decodes structural packets and drives dual L298N H-Bridge motor drivers to regulate the speed and direction of all four wheels.



3. Transmitter Hardware Components

ESP32 Development Board
Features a 32-bit dual-core processor running at 240 MHz. Handles sensor initialization, complementary filtering, and ESP-NOW broadcasts.

MPU6050 6-Axis Motion Sensor
Combines a 3-axis accelerometer and a 3-axis gyroscope on a single silicon die. Transmits orientation vectors over the hardware I2C bus.

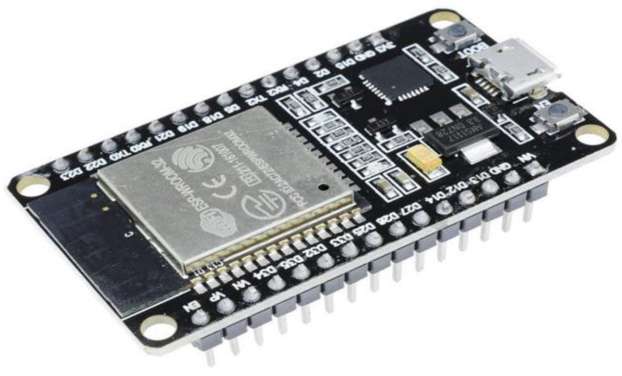
Physical Interface Connections
Operates at 3.3V. SCL is mapped to GPIO 22 and SDA is mapped to GPIO 21 on the transmitter board.

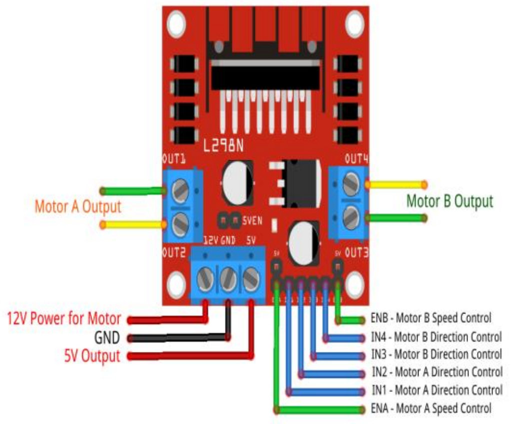
4. Receiver Hardware Components

ESP32 Receiver Board
Runs a dedicated ESP-NOW receiver callback routine. Decodes incoming packets and outputs logic commands and PWM values.

Dual L298N H-Bridge Drivers
Split into Front and Back configurations. Handles up to 2A per channel to power the motors without overloading the microcontrollers.

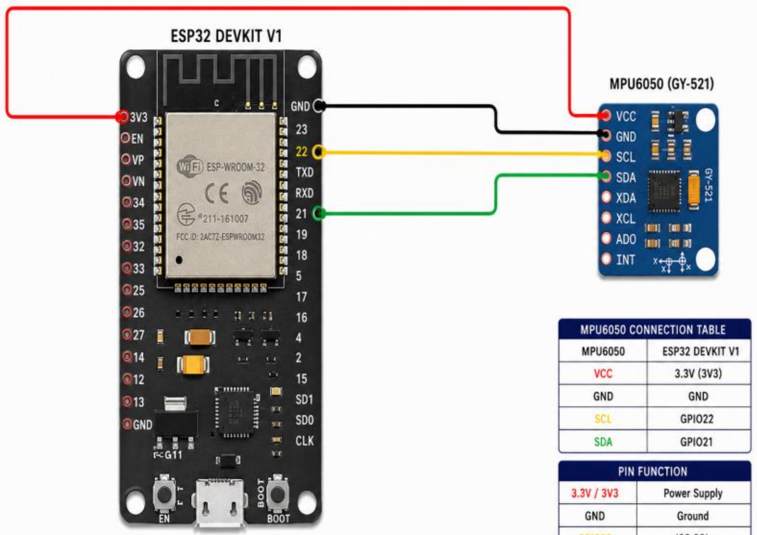
Actuators & 7.4V Power Supply
Four 200 RPM dual shaft BO geared motors provide optimal torque, driven by two series-wired 18650 Lithium-Ion batteries.





5. Circuit Diagram and Pin Configuration

1. TRANSMITTER (ESP32 + MPU6050)



MPU6050 CONNECTION TABLE

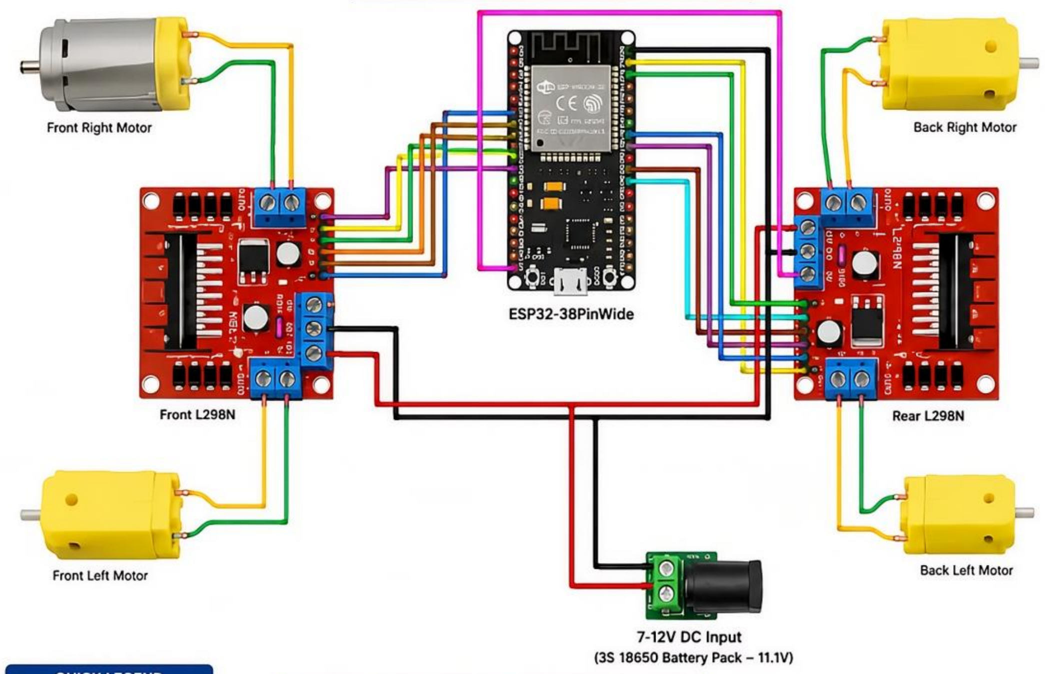
MPU6050	ESP32 DEVKIT V1
VCC	3.3V (3V3)
GND	GND
SCL	GPIO22
SDA	GPIO21

PIN FUNCTION

3.3V / 3V3	Power Supply
GND	Ground
GPIO22	I2C SCL
GPIO21	I2C SDA

ESP32 4WD CAR – RECEIVER CIRCUIT DIAGRAM

ESP32 + Dual L298N Motor Driver Wiring



QUICK LEGEND

- +12V / VIN
- GND
- Motor Wire
- Signal Wires (GPIO)

NOTES

- ESP32 is powered via VIN pin from the battery pack.
- All grounds must be common between ESP32, both L298N boards, and battery.
- This receiver circuit controls all 4 motors (4WD).
- Use proper battery and wire capacity for stable performance.

1. RECEIVER (ESP32) PIN CONFIGURATION

Function	ESP32 Pin	Connected To
IN1 (Front Left)	GPIO13	Front L298N IN1
IN2 (Front Left)	GPIO12	Front L298N IN2
IN3 (Front Right)	GPIO14	Front L298N IN3
IN4 (Front Right)	GPIO27	Front L298N IN4
ENA (Front Enable)	GPIO33	Front L298N ENA
ENB (Front Enable)	GPIO25	Front L298N ENB
IN1 (Back Left)	GPIO18	Rear L298N IN1
IN2 (Back Left)	GPIO19	Rear L298N IN2
IN3 (Back Right)	GPIO23	Rear L298N IN3
IN4 (Back Right)	GPIO26	Rear L298N IN4
ENA (Rear Enable)	GPIO32	Rear L298N ENA
ENB (Rear Enable)	GPIO15	Rear L298N ENB

2. MOTOR CONNECTIONS

Front L298N (Motors)			Rear L298N (Motors)		
Motor	Wire Color	L298N Terminal	Motor	Wire Color	L298N Terminal
Front Right	Black	OUT1	Back Right	Black	OUT1
Front Right	Red	OUT2	Back Right	Red	OUT2
Front Left	Red	OUT3	Back Left	Red	OUT3
Front Left	Black	OUT4	Back Left	Black	OUT4

3. POWER CONNECTIONS

Power (+) Connections

Battery Pack (+11.1V)
- → Front L298N +12V
- → Rear L298N +12V
- → ESP32 VIN

Power (–) Ground Connections

Battery Pack (–)
- → Front L298N GND
- → Rear L298N GND
- → ESP32 GND

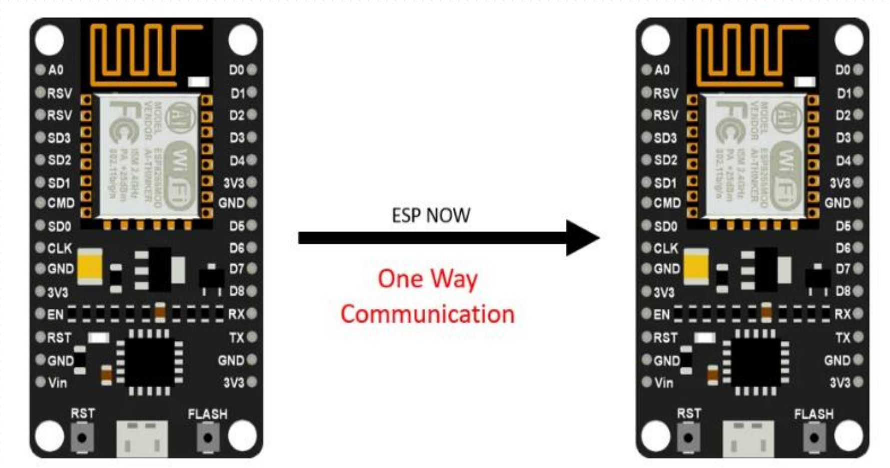
4. ENSURE THE FOLLOWING

- ☑ All grounds (GND) are connected together.
- ☑ ENA and ENB jumpers are removed from both L298N boards.
- ☑ Correct motor wires are connected to the correct outputs.
- ☑ Use a stable 7-12V DC power supply (3S 18650 recommended).

6. WHY ESP-NOW?

Why We Chose ESP-NOW :-

• Instant response.
• Low latency.
• Reliable communication.
• Simple implementation.
• Ideal for gesture-controlled robotic vehicles.



Advantages over Wi-Fi :-

• No Wi-Fi router or Internet connection required.
• Direct communication between two ESP32 boards.
• Faster response with very low latency.
• Easy pairing using the receiver's MAC address.
• Lower power consumption.

Advantages over Bluetooth:-

• Longer communication range.
• More stable communication.
• Faster data transmission.
• Lower communication delay.
• Better suited for real-time robotic control.

7. Software Implementation

Libraries Used :-

Transmitter
- WiFi.h
- esp_now.h
- Wire.h
- MPU6050_tockn.h

Receiver
- WiFi.h
- esp_now.h

Main Functions:- Transmitter
- Initialize MPU6050 sensor.
- Read Pitch and Roll angles.
- Detect hand gestures.
- Convert gestures into movement commands.
- Send commands using ESP-NOW.

Main Functions :- Receiver
- Receive commands from transmitter.
- Decode received command.
- Generate PWM signals.
- Control motor direction.
- Drive the motors accordingly.

Transmitter (Main logic):-

```
void loop()
{
  mpu.update();

  float pitch = mpu.getAngleX();
  float roll  = mpu.getAngleY();

  if(pitch > 20)            data.command = 'F';
  else if(pitch < -20)      data.command = 'B';
  else if(roll < -20)       data.command = 'L';
  else if(roll > 20)        data.command = 'R';
  else if(pitch > 20 && roll < -20) data.command = 'Q';
  else if(pitch > 20 && roll > 20)  data.command = 'E';
  else if(pitch < -20 && roll < -20) data.command = 'Z';
  else if(pitch < -20 && roll > 20)  data.command = 'C';
  else if(roll < -45)       data.command = 'A';
  else if(roll > 45)        data.command = 'D';
  else                      data.command = 'S';

  esp_now_send(receiverMAC,
               (uint8_t*)&data,
               sizeof(data));

  delay(20);
}
```

8. Gesture Commands

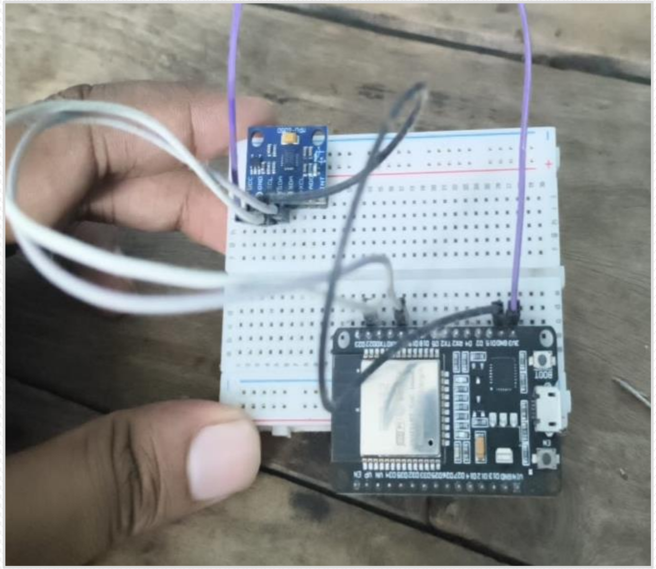
Gesture	Command	Car Movement
Hand Forward	F	Forward
Hand Backward	B	Backward
Hand Left	L	Left Strafe
Hand Right	R	Right Strafe
Forward Left	Q	Diagonal Forward Left
Forward Right	E	Diagonal Forward Right
Backward Left	Z	Diagonal Backward Left
Backward Right	C	Diagonal Backward Right
Strong Left Tilt	A	Rotate Left
Strong Right Tilt	D	Rotate Right
Flat	S	Stop

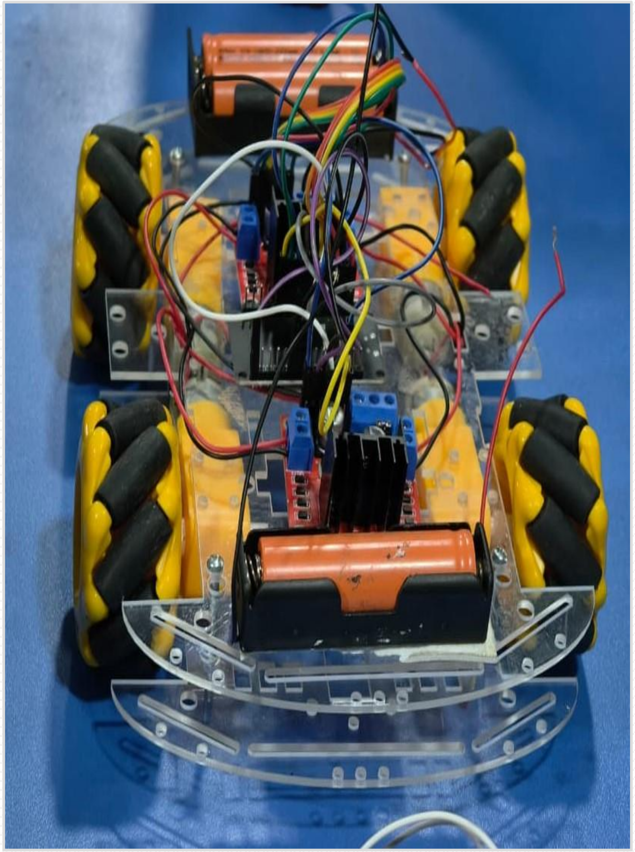
Receiver (Main logic):-

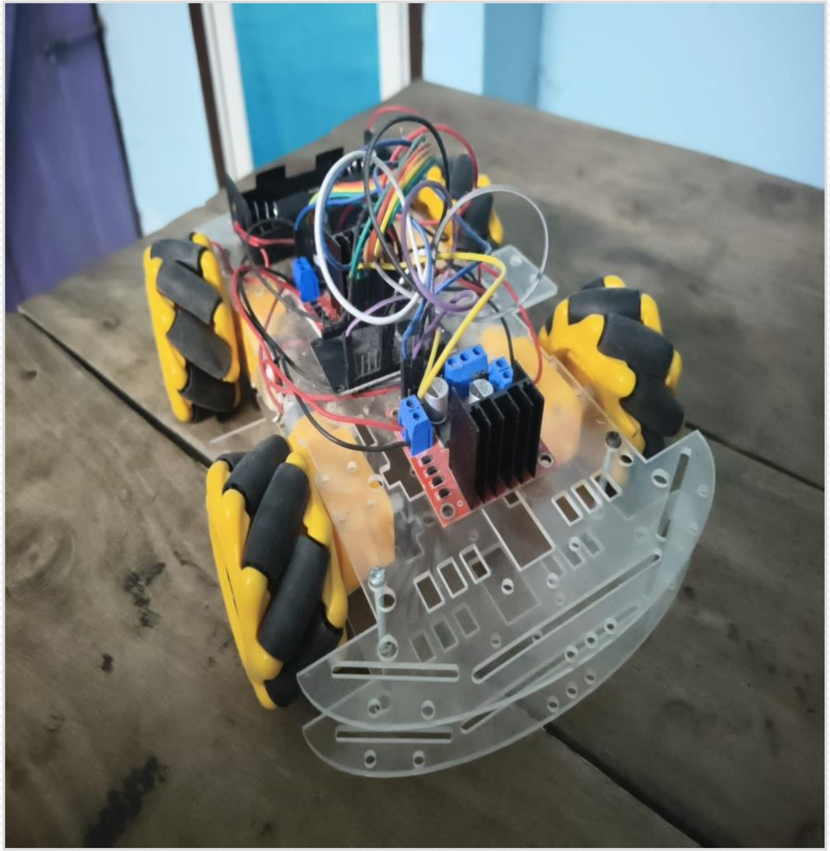
```
void onReceive(const esp_now_recv_info_t *info,
               const uint8_t *incomingData,
               int len)
{
  memcpy(&receiverData, incomingData, sizeof(receiverData));

  switch(receiverData.command)
  {
    case 'F': processCarMovement(FORWARD); break;
    case 'B': processCarMovement(BACKWARD); break;
    case 'L': processCarMovement(LEFT); break;
    case 'R': processCarMovement(RIGHT); break;
    case 'Q': processCarMovement(FORWARD_LEFT); break;
    case 'E': processCarMovement(FORWARD_RIGHT); break;
    case 'Z': processCarMovement(BACKWARD_LEFT); break;
    case 'C': processCarMovement(BACKWARD_RIGHT); break;
    case 'A': processCarMovement(TURN_LEFT); break;
    case 'D': processCarMovement(TURN_RIGHT); break;
    default : processCarMovement(STOP); break;
  }
}
```

9. Physical Assembly





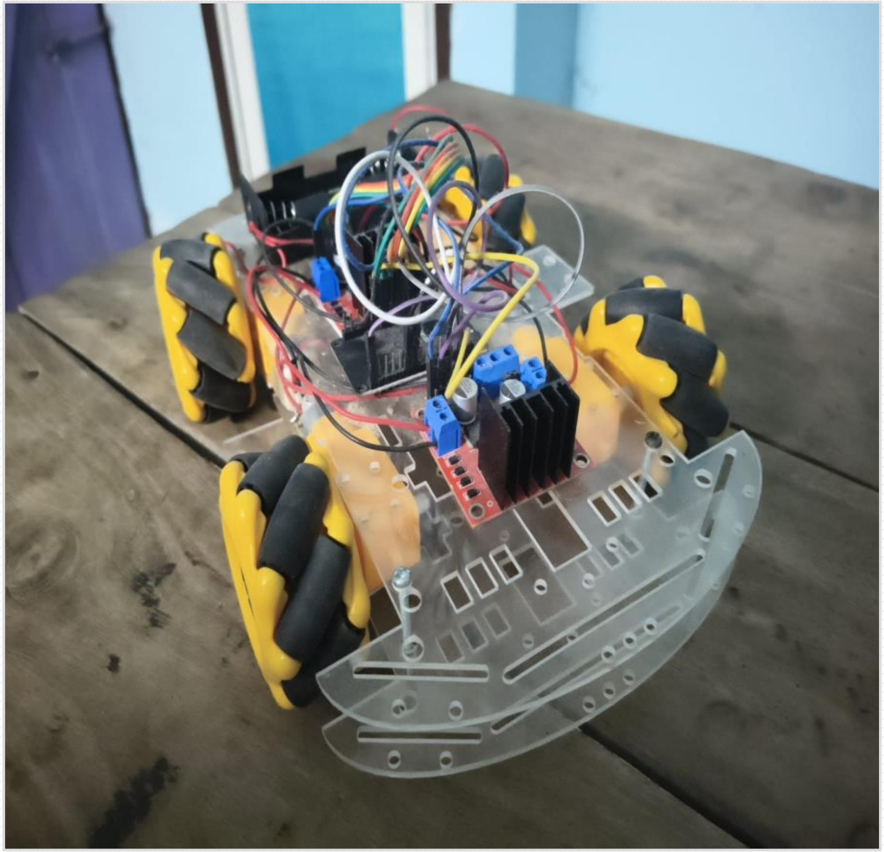


10. Performance Testing & Evaluation

Signal Latency Metrics
Achieves a highly responsive latency of 8 ms from hand tilt to motor reaction.

Wireless Range Testing
Maintains a stable, error-free line-of-sight wireless control range of up to 45 meters.

Operational Battery Runtime
Two series 18650 Li-Ion cells supply 30 minutes of continuous run time under full load.

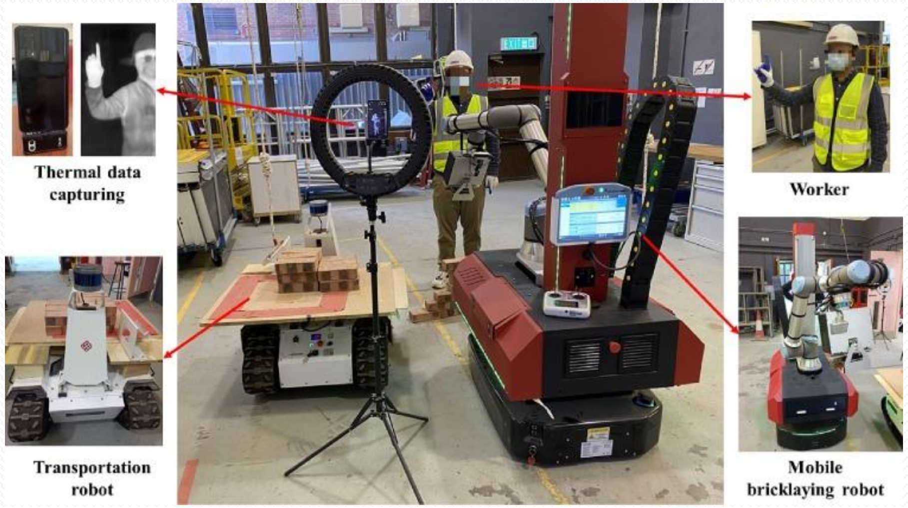


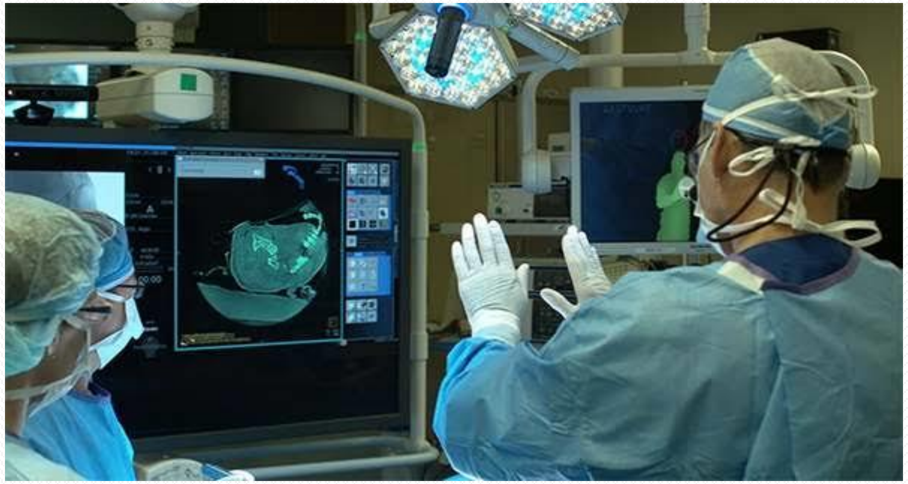
11. Applications

Medical Field
•Remote robotic surgery.
•Contactless operation in hospitals.
•Medical equipment control.
•Patient assistance robots.

Industrial Automation
•Material handling.
•Warehouse transportation.
•Factory automation.

Military & Defense
•Bomb disposal robots.
•Border surveillance.
•Hazardous area inspection.





Conclusions & Future Scope

Successful System Realization
Realized a low-latency connectionless ESP-NOW mobile robotic vehicle driven by wearable MPU6050 accelerometer wrist tilt gestures, providing a highly intuitive and responsive HMI interface.

Stable Physical Verification
System calibration and trials verified instant directional response (Forward, Reverse, Lateral Left/Right, Diagonal Drift) with high noise immunity and battery efficiency.

Future Research and Enhancements
Planned expansions include replacing L298N drivers with high-efficiency TB6612FNG drivers, integrating ultrasonic and LiDAR sensors for autonomous obstacle avoidance, implementing Kalman filters for sensor fusion, and integrating with the Robot Operating System (ROS) for warehouse routing.

Thank You